Coverage-Guaranteed Forecasting at Unseen Sensor Locations under Dynamic Topology

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***Abstract*—Spatiotemporal graph convolutional networks are tied to a fixed adjacency matrix, so they cannot produce any output at a sensor absent from training. We present OmniST, a forecaster whose computation has no structural dependence on the number of sensors and which, composed with inductive split-conformal calibration, returns distribution-free, coverage-guaranteed prediction intervals at never-seen sensor locations, keeping that guarantee as sensors are added or removed. A locality property explains why: each node's nonconformity score is exactly invariant to non-neighbor nodes, so the guarantee is recalibration-free under far-field topology change. Across four real datasets and three physical domains, coverage stays at or above the 90% target at unseen nodes under 30–50% node dropout, where four model-free interpolators each undercover on multiple domains, and OmniST also beats a learned inductive-kriging GNN. This is a condensed version of the full preprint (arXiv:2611.04256); auxiliary studies and the temporal-drift extension appear there.**

***Index Terms*—spatiotemporal forecasting, conformal prediction, inductive prediction, distribution-free uncertainty, dynamic graphs, sensor networks.**

I. Introduction

Sensor networks change: stations are added, fail, or go offline for maintenance. A standard spatiotemporal graph convolutional network (STGCN) [1] stores an $N \times N$ adjacency matrix and binds its weights to a specific node count, so when the network changes it needs an adjacency update or full retraining, and at a brand-new location it cannot predict at all. Beyond prediction, operators need to know how much to trust a forecast at such a location, which calls for a calibrated interval with a validity guarantee.

We present OmniST, a forecaster in which no component references a global node count. Composed with inductive split-conformal calibration, it produces distribution-free prediction intervals with a finite-sample coverage guarantee at unseen coordinates, and keeps the guarantee under node addition and removal. The contribution is this capability, not a point-accuracy record: on the standard fixed-topology benchmark OmniST matches rather than beats a comparable model.

II. Method

A. Topology-invariant architecture

OmniST obeys one invariant: for any component f and node i, $f(i)$ depends only on node i's own data and a fixed-size neighborhood $\mathcal{N}_k(i)$, never on the full node set. Four stages realize it. (1) *Coordinate tokenization*: each observation becomes a token $\mathbf{z}_i(t) = f_{\text{MLP}}(c_i) + f_{\text{time}}(t) + W_f \mathbf{x}_i(t)$, where a learned MLP maps a sensor's coordinates c_i to its spatial embedding, so a never-seen coordinate yields a valid token. (2) *k-NN attention*: spatial context is gathered from the k geographically nearest nodes, selected by Haversine distance at each forward pass, with multi-head attention and a learned distance bias; there is no $N \times N$ matrix, so the neighbor set adapts to whichever sensors are present. (3) *Per-node temporal processing*: a bidirectional GRU runs independently per node, so adding or removing a node changes no other node's representation. (4) A 2-layer MLP head maps to the forecast horizon.

B. Guaranteed-coverage inductive calibration

We wrap OmniST in split conformal prediction [2], [3]. With nonconformity score $s_i = |\hat{x}_i - x_i|$ and calibration set $\mathcal{C}$ of size n, the half-width is $\hat{q} = \text{Quantile}(\{s_i\}_{i \in \mathcal{C}}, \lceil (n+1)(1-\alpha) \rceil / n)$, and the interval is $[\hat{x} - \hat{q}, \hat{x} + \hat{q}]$. Under exchangeability this gives marginal coverage $\geq 1 - \alpha$. The inductive subtlety is that calibration residuals come from training stations while the test point is an unseen station, so the two score sets must stay exchangeable [4]. OmniST's per-node construction keeps the score distribution stable across node sets, which is what the next result formalizes.

C. Locality and recalibration-free coverage

Proposition 1 (Locality $\Rightarrow$ exact score invariance)**.** *Let $s_i = g(c_i, x_i, \{(c_j, x_j) : j \in \mathcal{N}_k(i; S)\}, y_i)$ be OmniST's nonconformity score for node i given the present node-set S, with $\mathcal{N}_k(i; S)$ its k nearest other nodes by Haversine distance. For any S' with $\mathcal{N}_k(i; S') = \mathcal{N}_k(i; S)$, in particular any addition or removal of nodes outside i's k-nearest set, $s_i(S') = s_i(S)$ exactly. Hence a split-conformal threshold $\hat{q}$ calibrated on training-station scores stays valid with no recalibration for any test node whose k-nearest set is unchanged.*

Proof sketch. Every component is a function of a node's own data and its k-NN receptive field only: the token depends on (c_i, x_i); the attention aggregates exactly $\mathcal{N}_k(i; S)$; the GRU and head act per node. No operation references N or a node index, and there is no cross-node reduction. So s_i is a fixed function of $(c_i, x_i, \mathcal{N}_k(i; S)\text{-data}, y_i)$, and if that argument is unchanged, so is s_i. Validity of $\hat{q}$ then follows from standard split conformal. □

This is stronger than the general inductive-GNN case, where message passing makes any node change perturb every embedding and the calibration scores drift. We verify it directly: dropping nodes outside a test node's k-NN set changes its prediction by $< 10^{-7}$ (bit-identical to float precision), whereas dropping a true neighbor changes it by $\sim 7 \times 10^{-3}$.

III. Experiments

We use four real datasets across three physical domains: Delhi (CPCB, 37 stations) and Beijing (UCI, 12) air quality, NOAA GHCN weather (30), and Caltrans PeMSD7-M traffic (228); the full paper adds PeMS-BAY (325) for scale. For each dataset we hold out stations, calibrate split conformal on training-station residuals, and measure coverage and width at the unseen stations (target 90%).

A. Inductive point prediction

At five held-out Delhi stations, OmniST's inductive MAE is 26.7 ± 1.2 μg/m^3, better than nearest-neighbor interpolation (30.3, about 12%) and than IGNNK [5], the canonical learned inductive-kriging GNN (34.0 ± 1.3, about 21%; our faithful reimplementation, not the authors' code). Point accuracy is not the headline, though: the decisive quantity is a calibrated interval at the unseen location, which neither baseline provides reliably (IGNNK under-covers at 0.89).

B. Guaranteed coverage under dynamic topology

Table I reports inductive coverage at unseen nodes on all four datasets, at full network and under 30–50% training-node dropout. OmniST holds the 90% target everywhere and keeps it under dropout. The nearest-neighbor baseline under-covers on three of the four. Against four model-free interpolators (nearest-neighbor, inverse-distance weighting, mean-of-k, and a kriging-style predictor), each wrapped in the same conformal procedure, OmniST is the only method that holds the target on all four datasets.

TABLE I
Inductive coverage at unseen nodes (target 90%).

Dataset (domain)	OmniST	NN	drop 30%	drop 50%
Delhi AQ (environmental)	0.914	0.925	0.912	0.916
Beijing AQ (environmental)	0.928	0.869	0.928	0.930
Weather (meteorology)	0.919	0.896	0.919	0.920
PeMSD7-M (transportation)	0.915	0.887	0.916	0.917

On Delhi, OmniST's intervals reach 0.915 coverage at a mean width of 128.5, 14% tighter than nearest-neighbor (0.925 coverage, width 149.7), which over-covers while carrying no finite-sample validity. Coverage stays near 0.91 after 30% or 50% of training stations are removed and the model is recalibrated on the survivors. An ablation (full paper) shows the coordinate-based construction, not the attention module or a particular k, is what earns the guarantee.

IV. Related Work

Fixed-topology STGCNs [1] and extreme-aware variants bind weights to a node count and cannot predict at an unseen node. Inductive kriging [5] predicts there but without a validity guarantee. A recent line attaches conformal guarantees to dynamic and temporal graphs [4], typically establishing validity as a graph's edges or labels evolve on a fixed node set. OmniST differs by targeting coverage at spatially unseen coordinates, under node addition and removal, and by showing the guarantee survives topology change across domains.

V. Limitations

The contribution is a capability, not a point-accuracy record; OmniST matches rather than beats a fixed-topology model in-distribution. The guarantee is marginal, not fully conditional. The learned baselines are faithful reimplementations, not the original code. The full paper reports a scale study to 325 real sensors, a temporal-drift extension that restores coverage over time with an online conformal layer, and complete multi-seed statistics.

Reproducibility

Code, data loaders, and a reproduction manifest are at \repourl. Full preprint: arXiv:2611.04256.

References

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